Chemistry - Graduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the mole fraction of H_2 SO_4 in a 7.0 molar solution of H_2 SO_4 which has a density of 1.39 g/ml?
   (a) 0.3
   (b) 0.7
   (c) 0.15
   (d) 0.9
   (e) 0.5

2. A certain radioactive material has a half-life of 36 minutes. Starting with 10.00 grams of this material, how many grams will be left after 2 hours?
   (a) $1.5 \times 10^-4 grams$
   (a) 2.50 grams
   (b) 3.33 grams
   (c) 4.00 grams
   (d) 1.00 grams

3. The rate law may be written using stoichiometric coefficients for which of the following?
   (a) Redox reactions
   (b) Elementary processes
   (c) Dissolution reactions
   (d) Polymerization reactions
   (e) Complexation reactions

4. Which of the following is expected to have two or more resonance structures?
   (a) CH 4
   (b) NH 3
   (c) CCl 2 F_2
   (d) H_2O
   (e) SF 6

5. Calculate the uncertainty ΔL_z for the hydrogen-atom stationary state: $2p_z$.
   (a) 1

(b) 0.75

(c) 2

(d) 4

(e) 0.5

True / False (10 marks)

Answer True or False

6. True or False: The molarity of sodium ions (Na^{+})$inhumanbloodserumisapproximately0.15M$.

6. True or False: The half-life of $^{3}3Pis40days, whichisconsistentwithitsdecaybehaviorinnuclearreactio$

6. True or False: The decomposition of KClO 3 to produce 0.96 g of oxygen requires 4.0 g of KClO 3.

7. True or False: Lead is a liquid element that is dark-colored and nonconducting at room temperature.

8. True or False: The pOH of a 0.0001 N KOH solution is 4, suggesting it is a neutral solution.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given an unknown element X with an atomic weight of 210.197 amu and isotopes $^{2}10Xand^{2}12Xwithmassesof209.64amuand211.66amu, calculatetheirrelativeabundancesanddiscuss$

11. In a series of two electrolytic cells containing AgNO 3 and CuSO 4, where 1.273 g of silver is deposited, analyze the relationship between the deposited silver and the amount of copper deposited concurrently. Calculate the mass of copper deposited and discuss the underlying principles of electrolysis that support your findings.

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